

Q1 Palindrome number/string

10 marks

Palindrome number/string

Your Code

```
1 class Main{
2     public static void main(String[] args){
3         int num = 121;
4         int temp = num;
5         int rev = 0;
6         while (temp>0){
7             rev = rev*10 + temp%10;
8             temp= temp/10;
9         }
10        if (num==rev)
11            System.out.println(num+ " is a pallindrome number");
12        else{
13            System.out.println(num+" is not an pallindrome number");
14        }
15    }
16 }
```

Input

stdin input

Output

121 is a pallindrome number

Q2 First n perfect numbers

10 marks

First n perfect numbers

Your Code

```
1 class Main{
2     public static void main(String[] args){
3         int n = 4;
4         int count = 0;
5         int num = 2;
6         while(count<n){
7             int sum = 0;
8             for(int i=1;i<num;i++){
9                 if (num%i==0){
10                    sum+=i;
11                }
12            }
13            if (sum==num){
14                System.out.println(num);
15                count++;
16            }
17            num++;
18        }
19    }
20 }
```

Input

stdin input

Output6
28
496
8128

Q3 Decimal number equivalent to binary number and octal number

10 marks

Decimal number equivalent to binary number and octal number

Your Code

```
1 public class Main{
2     public static void main(String[] args){
3         String binary = "1010";
4         String octal = "17";
5         int decimal1 = Integer.parseInt(binary, 2);
6         int decimal2 = Integer.parseInt(octal, 8);
7         System.out.println("Binary number = "+ binary);
8         System.out.println("Decimal Equivalent = "+decimal1);
9         System.out.println("Octal number = "+ octal);
10        System.out.println("Decimal Equivalent = "+decimal2);
11    }
12 }
```

Input

stdin input

Output

```
Binary number = 1010
Decimal Equivalent = 10
Octal number = 17
Decimal Equivalent = 15
```

Q4 Nth max number and nth min number in array then sum of it and difference of it

10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```
1 import java.util.Arrays;
2 public class Main{
3     public static void main(String[] args){
4         int[] arr = {7,2,9,4,1,6,8,3,5};
5         int n = 2;
6         Arrays.sort(arr);
7         int nthMin = arr[n-1];
8         int nthMax = arr[arr.length-n];
9         System.out.println("Nth Minimum =" + nthMin);
10        System.out.println("Nth Maximum =" + nthMax);
11        System.out.println("sum="+(nthMin+nthMax));
12        System.out.println("difference=" + (nthMax-nthMin));
13    }
14 }
```

Input

stdin input

Output

```
Nth Minimum =2
Nth Maximum =8
sum=10
difference=6
```

Q5 Find the average positive and negative numbers upto -1

10 marks

Find the average positive and negative numbers upto -1

Your Code

```
1 class Main {
2     public static void main(String[] args) {
3         int a[] = {10, 20, -5, 15, -10, -1};
4         int ps = 0, ns = 0, pc = 0, nc = 0;
5         for (int i = 0; i < a.length; i++) {
6             if (a[i] == -1)
7                 break;
8             if (a[i] > 0) {
9                 ps += a[i];
10                pc++;
11            } else {
12                ns += a[i];
13                nc++;
14            }
15        }
16        System.out.println("Positive Average = " + (double) ps / pc);
17        System.out.println("Negative Average = " + (double) ns / nc);
18    }
19 }
```

Input

stdin input

Output

```
Positive Average = 15.0
Negative Average = -7.5
```

Q6 Removing duplicates in array

10 marks

Removing duplicates in array

Your Code

```
1 class Main {
2     public static void main(String[] args) {
3         int[] a = {1, 2, 3, 2, 4, 1, 5};
4         System.out.println("Array after removing duplicates:");
5         for (int i = 0; i < a.length; i++) {
6             boolean duplicate = false;
7             for (int j = 0; j < i; j++) {
8                 if (a[i] == a[j]) {
9                     duplicate = true;
10                    break;
11                }
12            }
13            if (!duplicate) {
14                System.out.print(a[i] + " ");
15            }
16        }
17    }
18 }
```

Input

stdin input

Output

```
Array after removing duplicates:
1 2 3 4 5
```

Q7 Matrix multiplication

10 marks

Matrix multiplication

Your Code

```
1 class Main {
2     public static void main(String[] args) {
3         int a[][] = {{1, 2}, {3, 4}};
4         int b[][] = {{5, 6}, {7, 8}};
5         int c[][] = new int[2][2];
6         for (int i = 0; i < 2; i++) {
7             for (int j = 0; j < 2; j++) {
8                 for (int k = 0; k < 2; k++) {
9                     c[i][j] += a[i][k] * b[k][j];
10                }
11            }
12        }
13        System.out.println("Result:");
14        for (int i = 0; i < 2; i++) {
15            for (int j = 0; j < 2; j++) {
16                System.out.print(c[i][j] + " ");
17            }
18            System.out.println();
19        }
20    }
```

Input

stdin input

Output

```
Result:
19 22
43 50
```